Atlantic croaker (*Micropogonias undulates*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Low
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.28	2.35	
Complexity in Reproductive Strategy	2.48	1.9	
Dispersal of Early Life Stages	1.28	2	
Egg and Larval Survival and Settlement Requirements	2.92	1.7	
Historical Variability in Temperature	1.12	2.7	
Juvenile Habitat Requirements	2.24	2.9	
Other Stressors	2.68	2.1	
Population Growth Rate	1.56	1.8	
Prey Specificity	1.28	2.7	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.68	2.5	
Spawning Cycle	2.48	2.3	
Stock Size/Status	2.52	1	
Total Sensitivity | Certainty	**Moderate (2) | 0.9**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.02	0.94	
Bottom Oxygen	1.91	0.84	
Bottom Salinity	2.99	0.68	
Bottom Aragonite Solubility	1.22	0.59	
Net primary production	1.2	0.25	
Downward Flux of Particulate Organic Carbon	1.31	0.36	
Total Exposure	**Low (1.49)**	**0.78**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.97)**	**1.56**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic croaker. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Moderate	Increase
Exposure	Very High	Low	Decrease
Vulnerability	Moderate	Low	Decrease
Directional Effect	Positive	Neutral	Increase

Stocks

Atlantic croaker is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

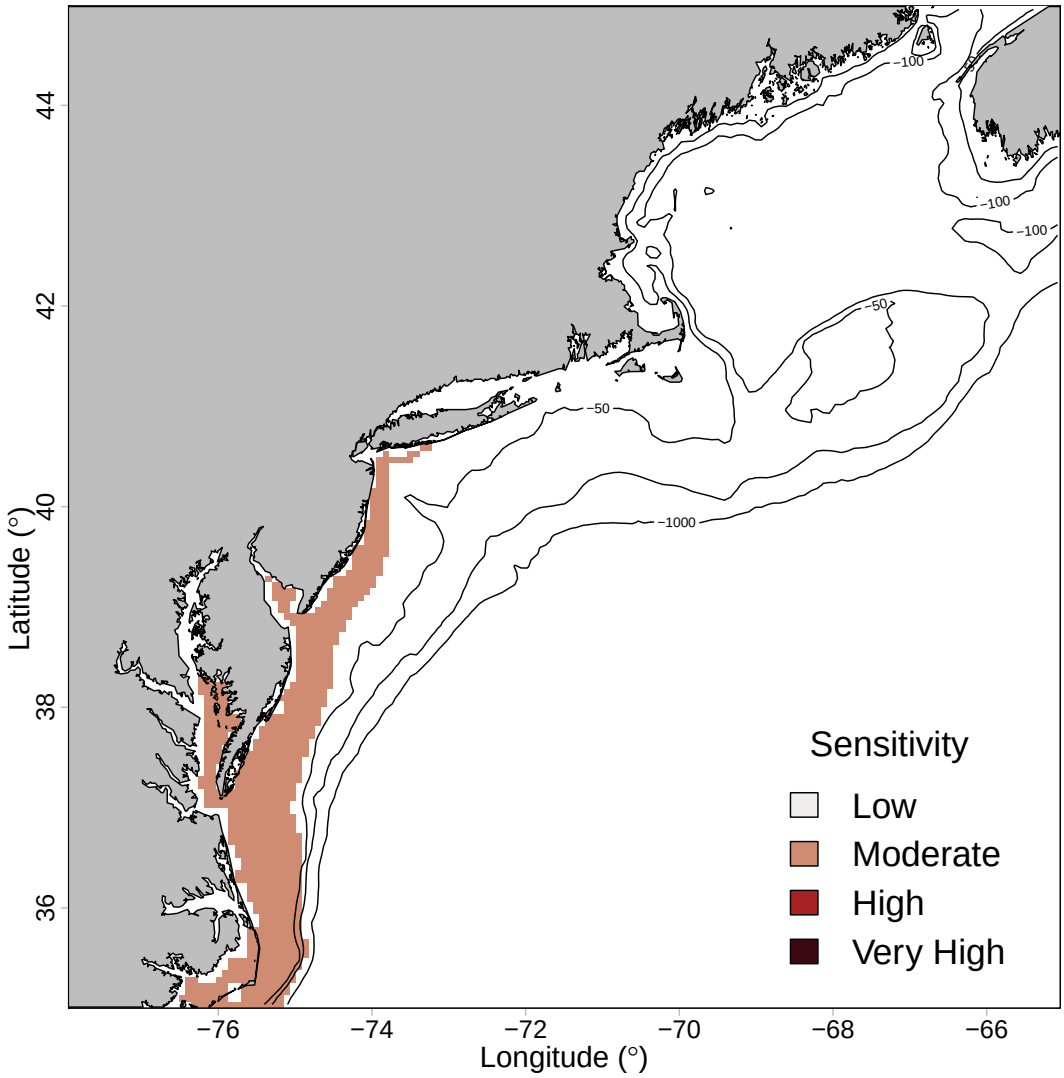


Figure 2: Atlantic croaker Sensitivity on the Northeast Shelf

Atlantic croaker received a sensitivity score of **Moderate** (Figure 2). Certainty was high (90%). Three attributes received scores greater than 2.5: 1) Egg and Larval Survival and Settlement Requirements (2.9), 2) Other Stressors (2.7), and 3) Stock Size (2.5) (Figure 1).

Croaker eggs and larvae rely on currents and tidal activity for transport to suitable nursery habitats (Sempsrott et al. 2017). Larvae are selective feeders, only feeding on a subset of plankton species (Govoni et al. 1986). Juveniles are also dependent on native salt marsh habitats, which themselves are highly vulnerable (Guyant et al. 2022) and are susceptible to eutrophication and hypoxic events.

Croaker had a low sensitivity score in the previous assessment (Hare et al. 2016). Egg and Larval Survival and Settlement Requirements drove the previous sensitivity score.

Exposure:

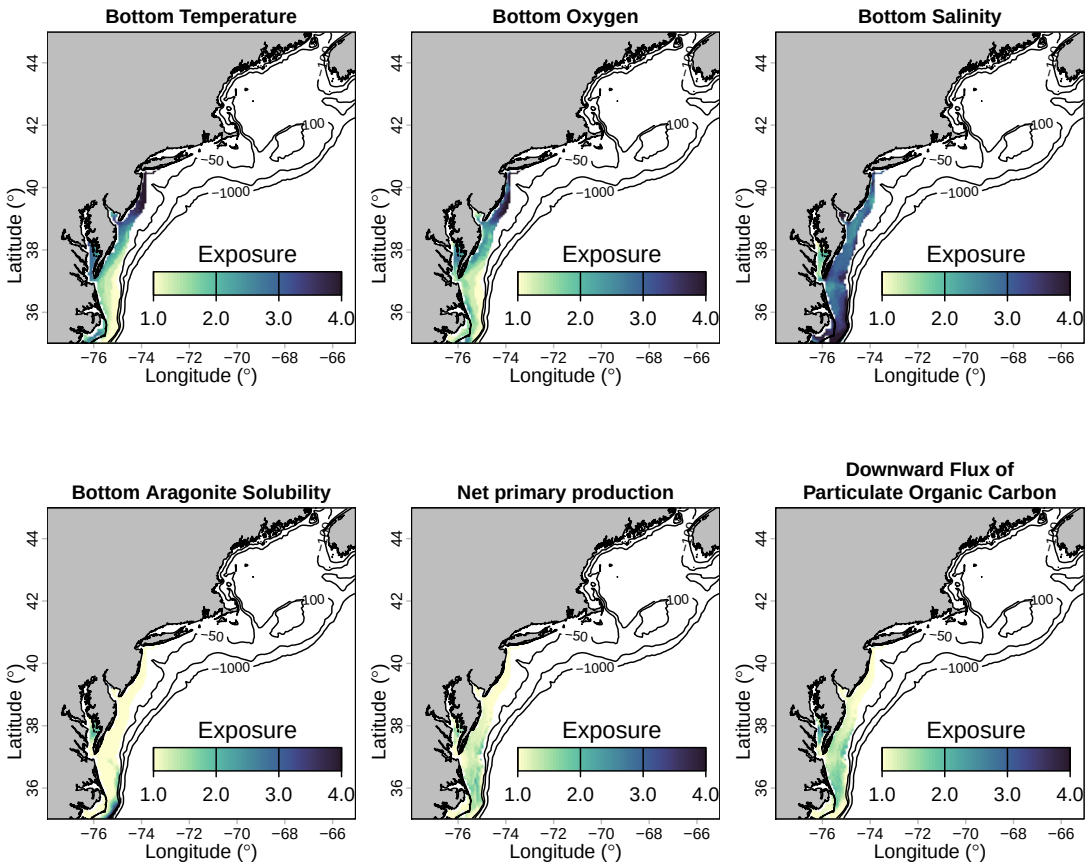


Figure 3: Maps of each Exposure Factor for Atlantic croaker. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure was **Low (1.49 +/- 0.78)** over the range of Atlantic croaker (Figure 1). Exposure was greatest along the coast of New Jersey, in Chesapeake Bay, and near the shelf break near Cape Hatteras (Figure 4). On the shelf and in the coastal bays, this high exposure was driven by bottom temperatures, oxygen concentrations, and bottom salinity. On the shelf break, high exposures were due to aragonite saturation state (Figure 3).

In the previous assessment, croaker received a Very High score, driven by high exposure to changes in sea surface temperature, air temperature, and ocean acidification (Hare et al. 2016).

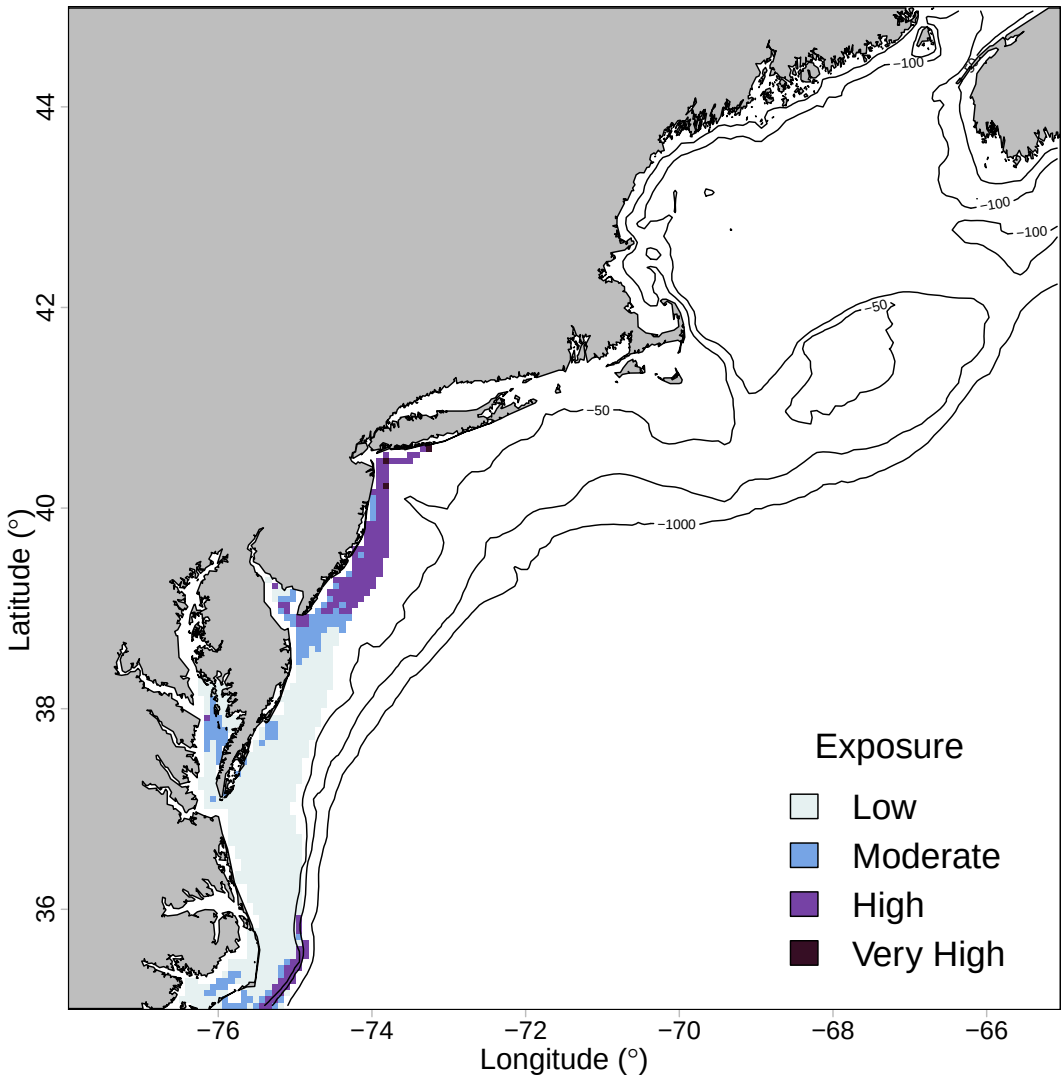


Figure 4: Total Exposure on the Northeast Shelf for Atlantic croaker. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

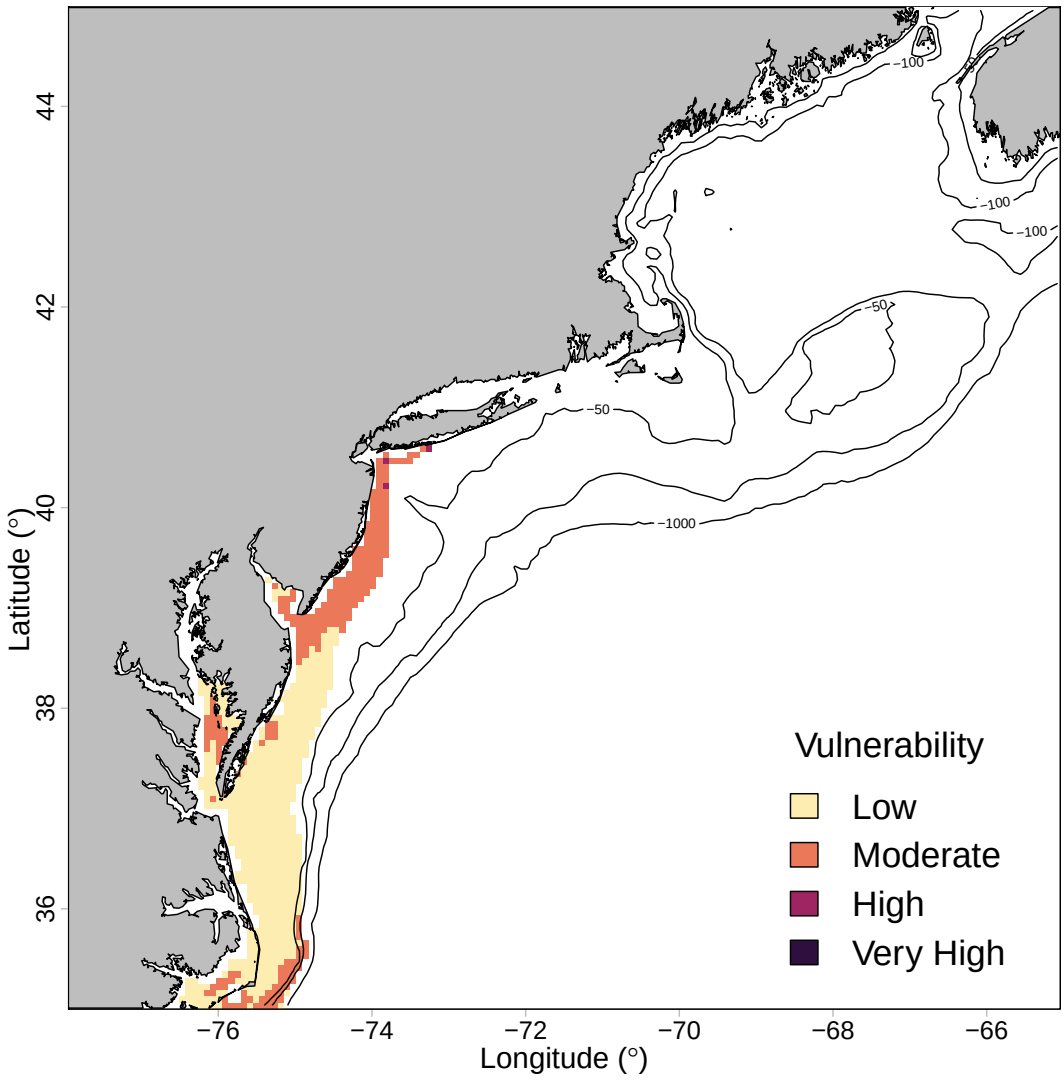


Figure 5: Vulnerability on the Northeast Shelf for Atlantic croaker. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability was **Low** for Atlantic croaker (2.97 +/- 1.56) (Figure 1). Scores were Moderate off the coast of New Jersey, in Delaware and Chesapeake Bays, and around the shelf break off the coast of Cape Hatteras. Scores and their distributions were similar when using the exposure scores calculated from only important factors (2.66 +/- 1.29) (Figure 5).

Atlantic croaker received a Moderate Vulnerability score in the previous assessment, driven by a Very High Exposure score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

73% of sensitivity attribute data quality scores were greater than or equal to 2, indicating good data quality. Four attributes were considered data gaps (mean attribute data quality score < 2): Complexity in Reproductive Strategy, Egg and Larval Survival and Settlement Requirements, Population Growth Rate, and Stock Size/Status (Figure 1).

Model Confidence Score:

The Atlantic croaker ensemble model received a model confidence score of 2.7 (+/- 0.4), indicating strong confidence in the model results. Experts stated that the model accurately represented croaker distributions, but may have underestimated presence in the winter months. The ensemble model's AUC was 0.99, indicating excellent performance.

Additional Metrics

Distribution Shifts:

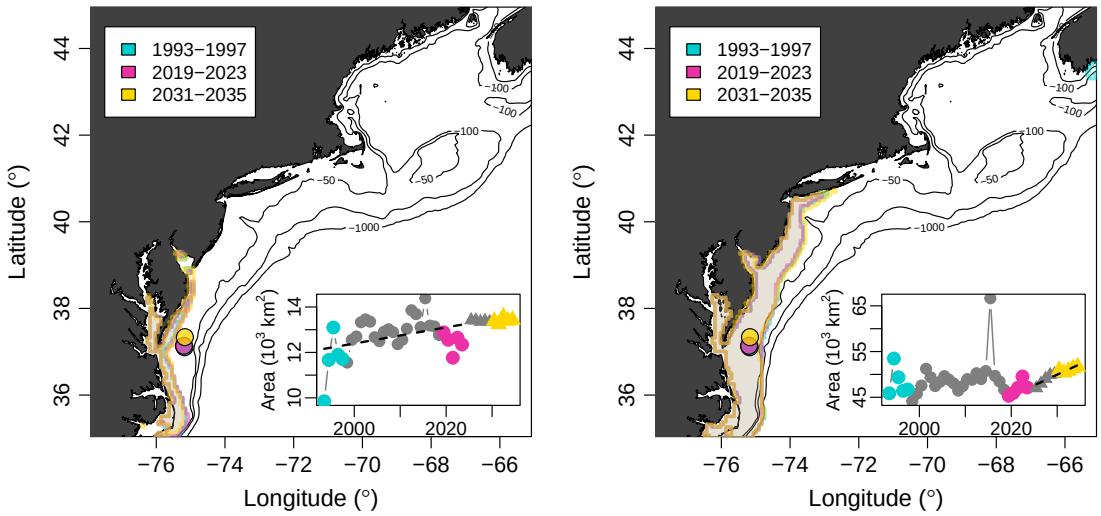


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic croaker. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Despite a significant positive trend, net core habitat growth was small between 1993 and 2023 (Figure 6). As a result, the weighted centroid shifted 4 km NW. Total habitat area remained consistent over the same time period.

Croaker habitats are predicted to expand in the near future. The distribution centroid is predicted to shift a further 22 km northward, driven by a northward range expansion. Croaker are predicted to occupy more space further offshore off New Jersey and further east along the coast of Long Island (Figure 6).

In the previous Assessment, croaker received a High Distributional Vulnerability rank due to their wide larval dispersal and migration behavior (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	11,766	49,314	4.55	2.33
2019-2023	4.2 km NW	12,504 (+737)	46,907 (-2,406)	4.49 (-0.06)	2.24 (-0.08)
2031-2035	22.3 km N	13,437 (+933)	51,029 (+4,121)	4.55 (+0.06)	2.33 (+ 0.08)

Directional Effect:

The effect of climate change on croaker is estimated to be **neutral** with 89% certainty in expert scores. Warmer temperatures may decrease young-of-year mortality rates. However, juvenile habitats are highly vulnerable to climate change. In the previous assessment, the effects of climate change were estimated to have a positive impact on croaker (Hare et al. 2016).

Climate Effects on Abundance and Distribution

As a sub-tropical species, Atlantic croaker may experience both positive and negative effects on their abundance and distribution as a result of climate change. Interannual variability in croaker abundance may be climate related, with colder winters causing increased mortality in overwintering young-of-the-year (Lankford Jr and Targett 2001). Warm winters, however, may result in increased abundances, or "outbursts" of juveniles in the spring that contribute to expansion of the population's distribution (Hare and Able 2007). Diamond, Murphy, and Rose (2013) modeled Atlantic croaker abundance in the Mid-Atlantic with global climate change projections and found increased croaker abundance with lower inter-annual variability under global climate change. Positive responses in sensitivity analyses stemmed from warmer temperatures, and negative responses were due to rising variability in salinity and increases in offshore transport and sea level rise (Diamond, Murphy, and Rose 2013). Common juvenile habitat types (submerged aquatic vegetation, salt marshes, and intertidal habitats) are highly vulnerable to climate change (Guyant et al. 2022).

Life History Synopsis

Atlantic croaker are a demersal species found from the Gulf of Maine to Argentina. Along the United States Atlantic coast, they are most abundant from Chesapeake Bay to northern Florida (Atlantic States Marine Fisheries Commission 2020). Atlantic croaker spawn from July - December in inshore shelf and estuarine waters between Delaware Bay and Cape Hatteras, peaking in August - October near Chesapeake Bay (Able and Fahay 1998). Their name comes from the croaking noises the fish make during spawning

by vibrating their abdominal muscles against their swim bladder (Atlantic States Marine Fisheries Commission 2020). After 30-90 days post-hatch Atlantic croaker larvae are driven inshore by winds (Thorrold, Jones, and Campana 1997) where they migrate to estuarine habitats (Sempsrott et al. 2017). Initial fall recruitment of croaker depends on fall continental shelf winds to provide transport into Chesapeake Bay (Norcross 1991). Recruitment to specific estuaries is influenced by tidal activity and currents, such that eggs and larvae are dependent on transport to suitable habitats for survival (Sempsrott et al. 2017). Atlantic croaker larvae are selective feeders, as they eat fewer taxa of suitable size than are present in ambient plankton (Govoni et al. 1986). Juvenile croaker are highly dependent on marine, estuarine, and riverine mud and sand substrates, as well as native salt marsh habitats (Guyant et al. 2022). The major prey of young-of-the-year Atlantic croaker are polychaetes, copepods, and mysids. Detritus is also a major component of the juvenile diet (Florida Fish and Wildlife Conservation Commission 2010; Sink 2011). Adult Atlantic croaker perform seasonal migrations, moving northward and inshore to estuaries and bays in the spring - fall and southward and offshore to shelf waters in the winter (Atlantic States Marine Fisheries Commission 2020). They reach maturity by 2 years of age and can live up to 17 years, although lifespans are commonly below 10 years (Atlantic States Marine Fisheries Commission 2020). Adult Atlantic croaker are opportunistic feeders and are known to consume polychaetes, anchovies, mysid shrimp, amphipods, fishes and crabs, and detritus in Chesapeake Bay (Nye, Loewensteiner, and Miller 2011). Atlantic croaker are managed under Amendment 1 to the Atlantic States Marine Fisheries Commission's Interstate Fishery Management Plan for Atlantic Croaker (2005) and Addenda I, II, and III (2011, 2014, 2020). (Atlantic States Marine Fisheries Commission 2020). The most recent benchmark stock assessment was completed in 2017, where the current stock status could not be determined because the assessment results were sensitive to certain modeling assumptions, but any further depletion in the stock was unlikely (Atlantic States Marine Fisheries Commission 2017).

Supplemental Scores

Exposure Timeseries

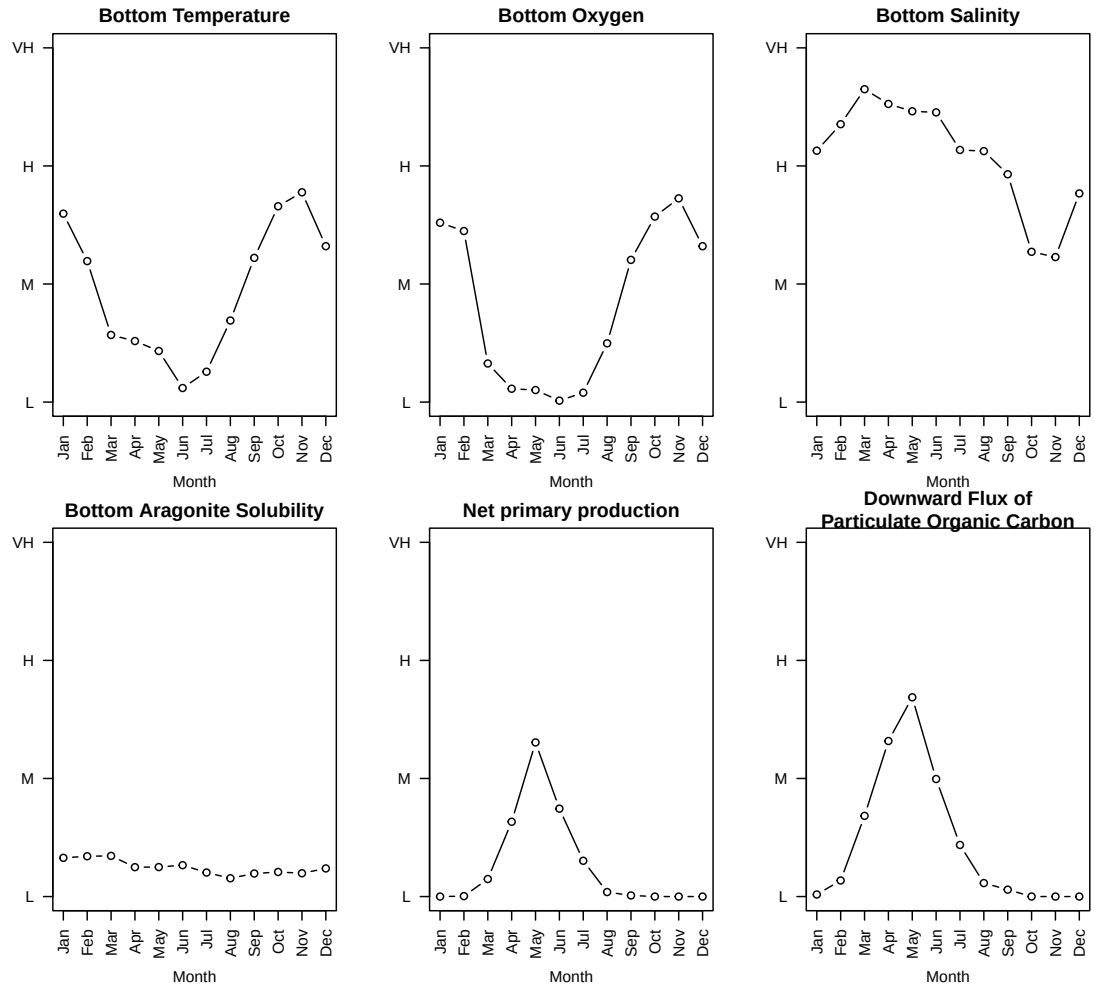


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic croaker. Exposure was calculated between 2014-2023 and 2025-2035

Across months, exposure ranged between Low and Moderate. Exposure was Low from February to April and June through September due to declines in bottom temperature, oxygen, and salinity exposure (Figure 8; Figure 7).

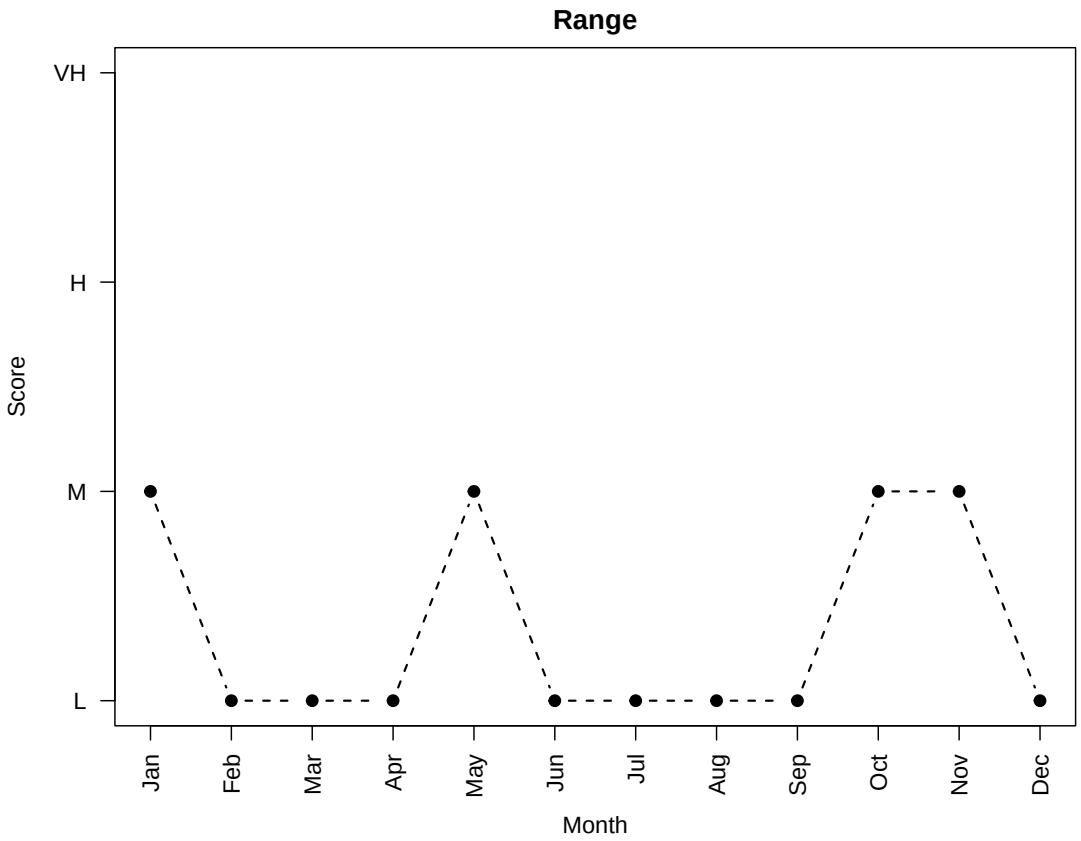


Figure 8: Timeseries of Total Exposure for Atlantic croaker

Exposure due to Significant Factors

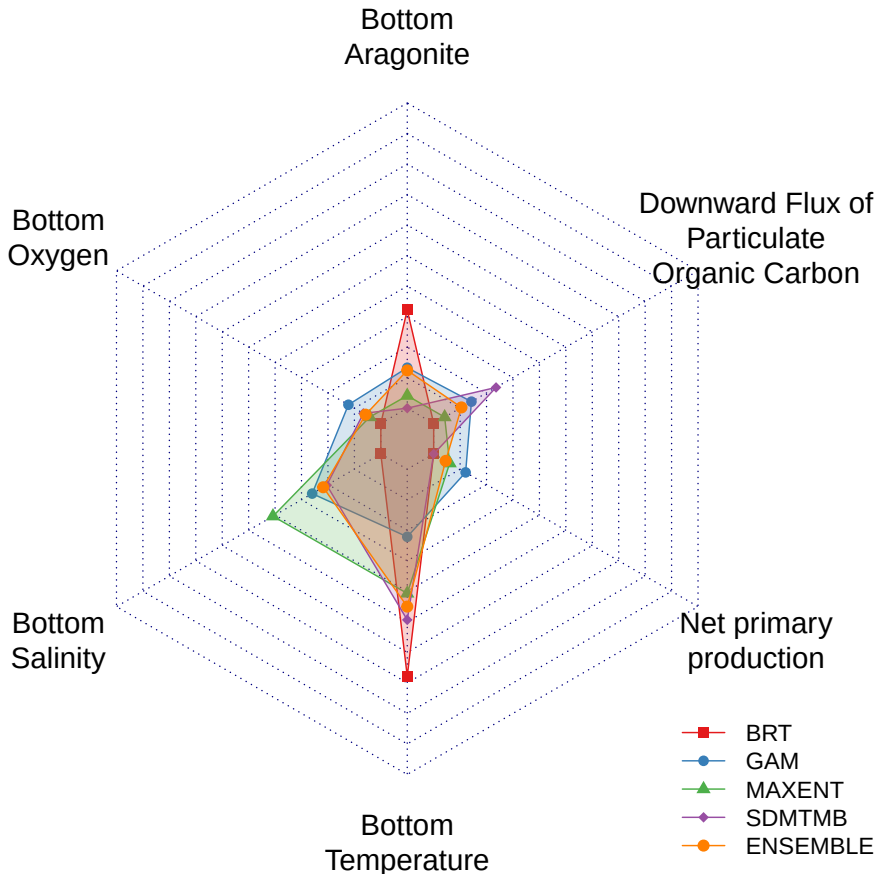


Figure 9: Dynamic Factors driving Atlantic croaker distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, salinity, and aragonite saturation state, in addition to integrated downward particulate organic carbon flux, were all important drivers of croaker distribution (Figure 9; Figure 11; Figure 10). Both adult and juvenile croaker use demersal estuarine habitats (Atlantic States Marine Fisheries Commission 2020; Guyant et al. 2022). Detritus is a major component of the juvenile diet (Florida Fish and Wildlife Conservation Commission 2010; Sink 2011) whereas adults are generalist feeders, consuming polychaetes, fishes, crabs, and detritus (Nye, Loewensteiner, and Miller 2011). Only including these factors in the exposure calculation did not alter the Exposure score (Figure 11).

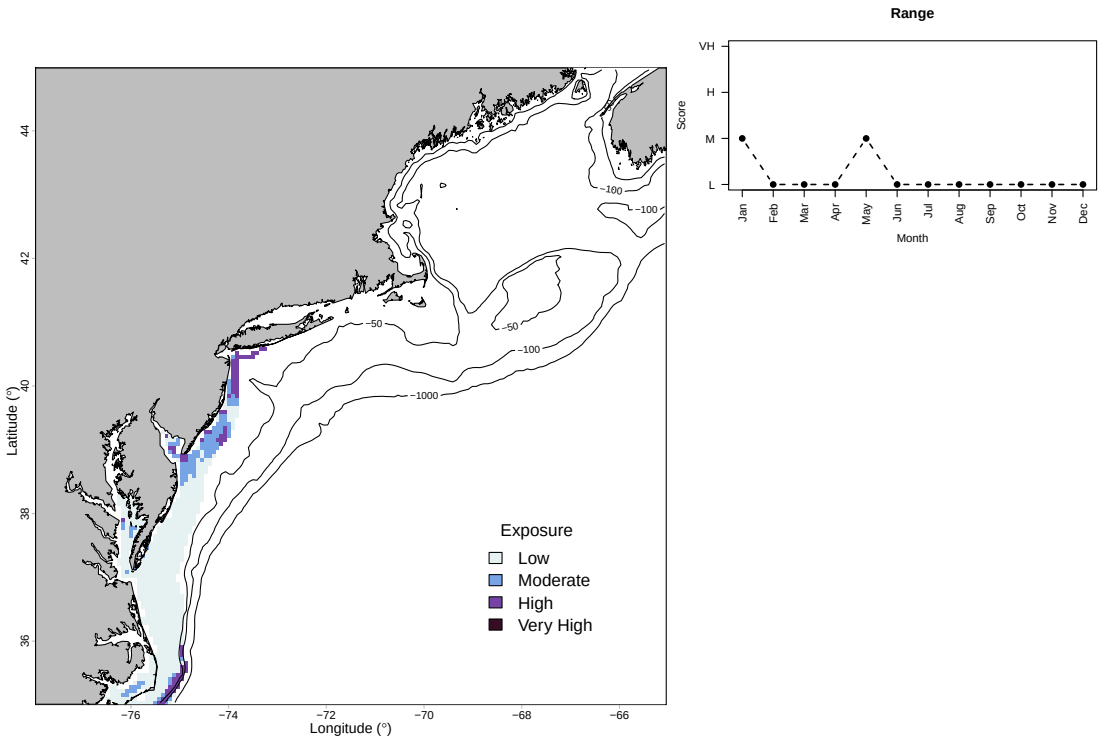


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic croaker distributions

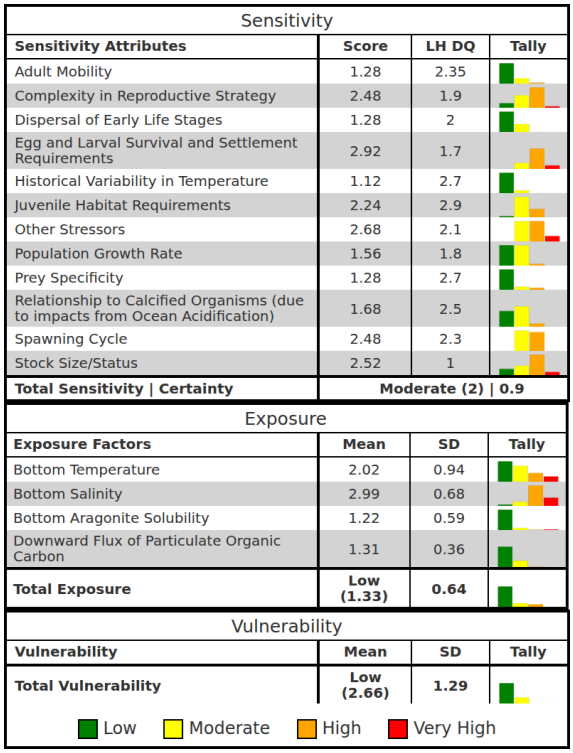


Figure 11: As in Figure 1 for Atlantic croaker, but with only significant Exposure factors and the resulting Vulnerability Scores.

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